Disadvantages

1. It cannot be used for amplifying high frequencies.
2. The operating point is shifted due to temperature variations.

Applications

Direct coupled amplifiers are used in television receivers, computers, regulator circuits and other electronics equipment.

5.11 VOLTAGE AMPLIFIER AND POWER AMPLIFIER

5.11.1 Voltage Amplifier.

A transistor amplifier which raises the voltage level (amplitude) of the signals is known as voltage amplifier

The first few stages in the multistage amplifier have the function of only voltage amplification. However, the last stage is designed to provide maximum power is known as power stage. Figure 5.25 shows the voltage amplifier and its use in audio circuits.

The first-stage voltage amplifier is called pre-amplifier. The next stage voltage amplifier is called driver amplifier.

5.11.2 Power Amplifier

A transistor amplifier which raises the power level of the signals is known as power amplifier.

A power amplifier is required to deliver a large amount of power and as such it has to handle large current. In order to achieve high power amplification, the following features are incorporated in such amplifiers.

1. The size of power transistor is made considerably larger
2. The base is made thicker to handle large currents
3. Transformer coupling is used for impedance matching

Voltage Gain: The voltage gain of an amplifier is given by

$$A_V = \frac{\text{Voltage of the output AC signal}}{\text{Voltage of the input AC signal}}$$

In common emitter circuits, voltage gain is calculated using

$$A_V = \beta \times \frac{R_L}{R_{in}}$$

Where β is current amplification factor, R_L is the collector load resistance and R_{in} is the resistance of base-emitter junction

Current Gain: Current gain of an amplifier is given by

$$A_I = \frac{\text{Output current}}{\text{Input current}},$$

Power Gain: Power gain of an amplifier is calculated by the formula

$$A_P = A_I \times A_V$$

Where Ap is the power gain, A_I is the current gain and A_V is the voltage gain.

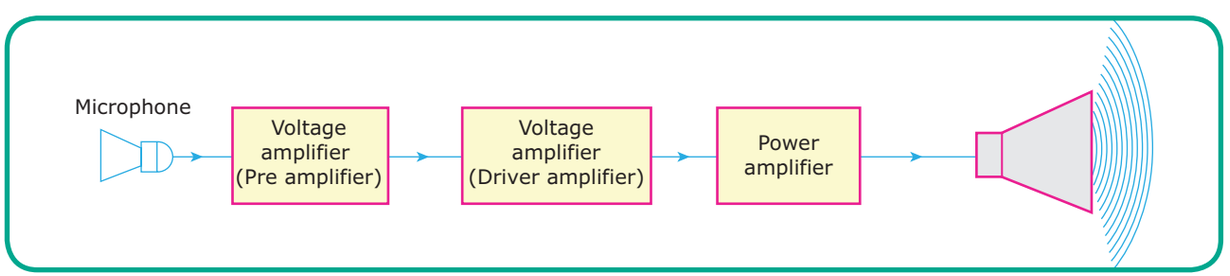


Figure 5.25 Voltage Amplifier

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